$$|\beta| \leqslant 33.75°$$

(i) $\alpha_1 = 30°$

a.

(2) $\sin\theta_1 = 2\sin\alpha_1 = 1$

$\theta_1 = 90°$

(3)

$$\theta_1' = \arcsin\left(\frac{2}{3}\right) = 41.81°$$

(4)

$$\beta_1 = 2\theta_1' - \theta_1 + \alpha_1 = 23.62° \quad (7)$$

$\beta_1 < \angle QOB$ (8)

b.

$i_C$

$$i_C = \arcsin\left(\frac{1.400}{1.500}\right) = 68.96° \quad (9)$$

$i = 90° - \alpha'$ $\alpha'$

$\alpha' \leqslant 90° - 68.96° = 21.04°$ (10)

$\alpha_1' = \theta_1 - \alpha_1 - \theta_1' = 18.19°$ (10)

(ii) $\alpha_2 = 25°$

(2) $\sin\theta_2 = 2\sin\alpha_2 = 2\sin25°$ $\theta_2 = 57.70°$

$\sin\theta_2' = \frac{1}{1.5}\sin\theta_2$ $\theta_2' = 34.30°$

$\beta_2 = 2\theta_2' - \theta_2 + \alpha_2 = 35.90°$ (11)

(6) $QQ'$

(iii) $\alpha_3 = 20°$

(2) $\sin\theta_3 = 2\sin\alpha_3 = 2\sin20°$ $\theta_3 = 43.16°$

$\sin\theta_3' = \frac{1}{1.5}\sin\theta_3$ $\theta_3' = 27.13°$

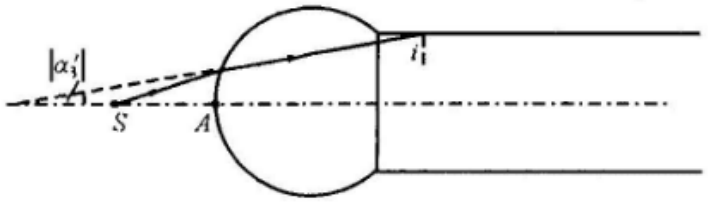
$\beta_3 = 2\theta_3' - \theta_3 + \alpha_3 = 31.10°$ (12)

(6) $QQ'$

$\alpha_3' = \theta_3 - \alpha_3 + \theta_3' = -3.97°$ (13)

$A$

4 $|\alpha_3'| < 21.04°$




4